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Professional Version

Atopic Dermatitis (Eczema)

(Atopic Eczema; Infantile Eczema; Neurodermatitis; Endogenous Eczema)

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Atopic dermatitis is a chronic relapsing inflammatory skin disorder with a complex pathogenesis involving genetic susceptibility, immunologic and epidermal barrier dysfunction, and environmental factors. Pruritus is a primary symptom; skin lesions range from mild erythema to severe lichenification to erythroderma.

(See also Definition of Dermatitis.)

Etiology of Atopic Dermatitis

Atopic dermatitis primarily affects children in urban areas or high-income countries, and prevalence has increased over the last 30 years; up to 20% of children and 10% of adults in high-income countries are affected (1, 2). While typically atopic dermatitis develops before age 5, and as early as 3 months of age, atopic dermatitis may also begin during late adulthood. An unproven hygiene hypothesis suggests that decreased early childhood exposure to infectious agents (ie, because of more rigorous hygiene regimens at home) may lead to an increase in autoimmunity to self-proteins and the development of atopic disorders (3).

Many patients or their family members who have atopic dermatitis also have allergic asthma and/or immediate-type hypersensitivities that manifest as allergic seasonal or perennial rhinoconjunctivitis. The triad of atopic dermatitis, allergic rhinoconjunctivitis, and asthma is called atopy or atopic diathesis.

Etiology references

1. Kantor R, Thyssen JP, Paller AS, Silverberg JI. Atopic dermatitis, atopic eczema, or eczema? A systematic review, meta-analysis, and recommendation for uniform use of 'atopic dermatitis'. *Allergy.* 2016 Oct;71(10):1480-5. doi: 10.1111/all.12982. Epub 2016 Aug 3. PMID: 27392131; PMCID: PMC5228598
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Pathophysiology of Atopic Dermatitis

The following factors contribute to the development of atopic dermatitis:

- Genetic factors
- Epidermal barrier dysfunction
- Immunologic mechanisms
- Environmental triggers

Genes implicated in atopic dermatitis are those encoding epidermal and immunologic proteins. In many patients, a major predisposing factor for atopic dermatitis is a loss-of-function mutation in the gene encoding for the filaggrin protein (1). Filaggrin, a component of the cornified cell envelope produced by differentiating keratinocytes, is critical to building the hygroscopic barrier of the stratum corneum (also called the natural moisturizing factor). Approximately 10% of European populations are heterozygous carriers of loss-of-function filaggrin mutations. The presence of these mutations (as well as intragenic copy mutations) increases the risk of more severe atopic dermatitis and higher IgE levels. Filaggrin mutations are also associated with peanut allergy and asthma, even in the absence of atopic dermatitis.

These molecular insights provide understanding of atopic dermatitis and how cutaneous inflammation, a T-cell–mediated delayed-type hypersensitivity, relates to allergic conditions with immediate-type hypersensitivities such as asthma and allergic rhinitis (hay fever). The epidermal skin barrier defect due to filaggrin mutations explains the development of xerosis and the predisposition to skin irritation. This results in the manifestations of atopic dermatitis, which is not an allergic reaction. The cutaneous inflammation, in contrast, is a T-cell–mediated delayed-type hypersensitivity and includes a Th2-dominant component in the skin. This hypersensitivity downregulates antimicrobial peptides (eg, beta-defensins), which explains why patients with atopic dermatitis are predisposed to developing bacterial and viral skin infections. The increased penetration of skin irritants and allergens also results in a Th2-dominant inflammation, which also drives IgE production and predisposes to immediate-type hypersensitivities. However, because the major mechanism mediating atopic dermatitis is delayed cell-mediated immunity, avoidance of immediate-type allergens (eg, pollen, dust mites) usually does not improve atopic dermatitis. Although immediate-type hypersensitivities (eg, asthma, allergic rhinitis) result from the skin barrier defect, they do not drive the T-cell–mediated cutaneous inflammation of atopic dermatitis.

In patients with atopic diathesis, atopic dermatitis typically precedes allergic rhinoconjunctivitis and asthma. Sometimes, this sequence is called the "atopic march" and occurs because the skin barrier defect is the primary deficiency in atopic conditions (2).

Pathophysiology references

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Symptoms and Signs of Atopic Dermatitis

Atopic dermatitis usually appears in infancy, as early as 3 months of age.

In the **acute phase,** lesions are intensely pruritic, red, thickened, scaly patches or plaques that may become eroded due to scratching.

In the **chronic phase,** scratching and rubbing create skin lesions that appear dry and lichenified.

Distribution of lesions is age-specific. In infants, lesions characteristically occur on the face, scalp, neck, eyelids, and extensor surfaces of the extremities. In older children and adults, lesions occur on flexural surfaces such as the neck and the antecubital and popliteal fossae.

Intense pruritus is a key feature. Itch often precedes lesions and worsens with dry air, sweating, local irritation, wearing wool garments, and emotional stress.

Common environmental triggers of symptoms include:

- Excessive bathing or washing
- Harsh soaps
- *Staphylococcus aureus* skin colonization
- Sweating
- Rough fabrics and wool

Individuals with atopic dermatitis often exhibit other dermatologic features of atopy. These may include xerosis, ichthyosis/palmar hyperlinearity (more prominent skin lines on the palms), keratosis pilaris, an infraorbital skinfold (Dennie-Morgan fold), thinning of the lateral eyebrows (Hertoghe sign), wool intolerance (irritation and itching triggered by skin contact with wool), white dermographism (vasoconstriction, causing whitening of the skin, in response to scratching), and increased transepidermal water loss (in nonaffected as well as affected skin).

Manifestations of Atopic Dermatitis

Flexural Atopic Dermatitis (1)

These 3 photos show erythema, scaling, and mild skin thickening on the antecubital fossae (left), wrists (center), and anterior ankles (right).

Photos courtesy of Thomas Ruenger, MD, PhD.

Flexural Atopic Dermatitis (2)

These 2 photos show erythema, scaling, and lichenification in the antecubital fossae (left) and popliteal fossae (right). In patients with dark skin tones, the erythema of dermatitis is often much less obvious and sometimes completely hidden by the skin pigmentation.

Photos courtesy of Thomas Ruenger, MD, PhD.

Atopic Dermatitis (Acute)

Atopic dermatitis usually develops in infancy. In the acute phase, lesions appear on the face and then spread to the neck, scalp, and extremities.

Image provided by Thomas Habif, MD.

Skin Lesion (Lichenification)

Lichenification is thickening of the skin with accentuation of normal skin markings; it is a result of chronic scratching or rubbing, which in this patient occurred during the chronic phase of atopic dermatitis.

Photo provided by Thomas Habif, MD.

Atopic Dermatitis (Chronic)

This photo shows extensive lichenification and hyperpigmentation of chronic atopic dermatitis secondary to long-term scratching.

Photo courtesy of Karen McKoy, MD.

Complications of atopic dermatitis

Secondary bacterial infections (superinfections), especially staphylococcal and streptococcal infections (eg, impetigo, cellulitis) are common.

Erythroderma (erythema that covers more than 70% of the body surface area) is rare but can result when atopic dermatitis is severe.

Acute Atopic Dermatitis with Secondary Infection

Hide Details

In this patient, acute dermatitis lesions were complicated by secondary infection (superinfection) with *Staphylococcus aureus* (impetiginization).

Impetiginized Atopic Dermatitis

IMAGE

This photo shows oozing and yellowish crusting on an erythematous background.

PHOTO COURTESY OF THOMAS RUENGER, MD, PHD.

Eczema herpeticum is an infection of the skin with herpes simplex virus (HSV) that is more diffuse and widespread than HSV skin infections in nonatopic patients. Eczema herpeticum is commonly the primary HSV infection in patients with atopic dermatitis, but can also result from recurrent infection. It manifests as grouped vesicles in areas of active or recent dermatitis, although normal skin can be involved. High fever and adenopathy may develop after several days. Occasionally, this infection can become systemic, which may be fatal. Sometimes the eye is involved, causing a painful corneal lesion. Eczema vaccinatum is a similar complication due to smallpox vaccination, in which the vaccinia virus disseminates and may become life-threatening. Atopic patients should therefore not receive smallpox vaccinations.

Eczema Herpeticum

Hide Details

This photo shows diffuse herpes simplex infection with grouped vesicles in a patient with atopic dermatitis.

Atopic patients are also prone to other viral skin infections (eg, common warts, molluscum contagiosum).

Patients with atopic dermatitis also have a higher risk of allergic contact reactions. For example, contact allergies to nickel, the most common contact allergen, are twice as common as in nonatopic patients. Frequent use of topical products exposes patients to many potential allergens, and allergic contact dermatitis caused by these products may complicate the treatment of atopic dermatitis.

Diagnosis of Atopic Dermatitis

- Clinical evaluation

Diagnosis of atopic dermatitis is clinical (1). History (eg, personal or family history of asthma or allergic rhinoconjunctivitis) is helpful. See table Diagnostic Clinical Features in Atopic Dermatitis for clinically relevant diagnostic features.

Atopic dermatitis is sometimes difficult to differentiate from other dermatoses. The diagnosis of atopic dermatitis depends on excluding other conditions, including:

- Seborrheic dermatitis
- Contact dermatitis (irritant or allergic)
- Psoriasis
- Nummular dermatitis (nonatopic)
- Scabies
- Ichthyoses
- Cutaneous T-cell lymphoma
- Photosensitivity dermatoses
- Immune deficiency diseases
- Erythroderma of other causes

A personal and/or family history of atopy and the distribution of lesions are helpful in making the diagnosis of atopic dermatits. The following distribution patterns can also help with differentiation:

- Psoriasis is typically easy to recognize by its sharply demarcated, thick, erythematous, and scaly plaques. It is usually extensoral rather than flexural and may include typical nail findings not common in atopic dermatitis (eg, oil spots, nail pits).
- Seborrheic dermatitis most commonly affects the face (eg, nasolabial folds, eyebrows, glabellar region, scalp).
- Nummular dermatitis is not flexural, and lichenification is rare. (However, coin-shaped, nummular plaques can occur in atopic dermatitis [nummular atopic dermatitis].)

Erythroderma caused by atopic dermatitis can be difficult to differentiate from erythroderma caused by other skin disorders.

Pearls & Pitfalls

- Clues to atopic dermatitis include a flexural distribution of lesions and personal or family history of allergic rhinoconjunctivitis and asthma.

TABLE

Diagnostic Clinical Features in Atopic Dermatitis*

Essential features
Pruritus
Dermatitis (eczema)—acute, subacute, or chronic, with

- Typical age-specific patterns†
- Chronic or relapsing history

Important features

Early age of onset

Personal or family history of atopic disease

IgE reactivity

Xerosis

Associated features (help to suggest the diagnosis)

White dermatographism

Keratosis pilaris

Pityriasis alba

Hyperlinear palms

Ichthyosis

Facial pallor

Periorbital changes

Perifollicular accentuation

Lichenification

Prurigo lesions

Delayed blanch response

Certain regional changes (eg, perioral changes, periauricular changes)

* Derived from: Guidelines of care for the management of atopic dermatitis: Section 1. Diagnosis and assessment of atopic dermatitis. *J Am Acad Dermatol.* 70(2):338–351, 2014. doi: 10.1016/j.jaad.2013.10.010.

† On the face, neck, and extensor surfaces in infants and children; flexural surfaces in any age group; spares groin and axillae.

There is no definitive laboratory test for atopic dermatitis. However, testing for type I allergies (prick, scratch, and intracutaneous testing or measurement of allergen-specific IgE levels and of total IgE levels) can help confirm an atopic diathesis. Cultures for *S. aureus* are not done routinely but can help determine sensitivities to systemic antimicrobials when impetiginization is suspected (eg, presence of yellowish crusts).

Diagnosis reference

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Treatment of Atopic Dermatitis

- Supportive care (including counseling on appropriate skin care and avoidance of precipitating factors)
- Antipruritics
- Topical corticosteroids
- Topical calcineurin inhibitors
- Topical crisaborole
- Topical Janus kinase (JAK) inhibitor (eg, ruxolitinib)
- Phototherapy, particularly narrow-band ultraviolet B
- Systemic immunosuppressants for moderate to severe disease
- Treatment of superinfections

Treatment of atopic dermatitis is most effective when addressing the underlying pathophysiologic processes.

Counseling on appropriate skin care and avoidance of triggers helps patients address the underlying skin barrier defect. Scratching lesions commonly increases itching and thus induces more scratching. Breaking this itch–scratch cycle is important.

Inflammatory flare-ups can be curtailed with topical immunosuppressants; phototherapy; and, if necessary; systemic immunosuppressants (1, 2, 3, 4, 5, 6).

Almost all patients with atopic dermatitis can be treated as outpatients, but patients who have severe superinfections or erythroderma may need to be hospitalized.

Supportive care

General skin care should be focused on the most common sources of skin irritation—excessive washing and harsh soaps:

- Limit the frequency and length of washing and bathing (showers/baths should be limited to once daily; sponge baths can be substituted to decrease the number of days with full baths)
- Limit the temperature of bathing water to lukewarm
- Avoid excessive rubbing; instead, pat skin dry after showering/bathing
- Apply moisturizers (ointments or creams—ceramide-containing products are particularly useful)
- For skin superinfection (eg, when yellowish crusting suggests impetigo), bathe with diluted bleach (eg, 60 mL [¼ cup] of household bleach [6%] into 76 L [20 gallons] of warm water)

Oral antihistamines can help relieve pruritus via their sedating properties. Options include hydroxyzine and diphenhydramine, preferably at bedtime to avoid the sedating effects during the day. Low-sedating or nonsedating antihistamines, such as loratadine, fexofenadine, or cetirizine may be useful, but their efficacy has not been established. Doxepin (a tricyclic antidepressant with H1 and H2 receptor blocking activity) may also help, but it is not recommended for children < 12 years.

Reduction of emotional stress is useful and helps break the itch–scratch cycle. Stress can affect the family (eg, being kept awake by a crying baby) as well as the patient (eg, being unable to sleep because of itching).

Dietary changes are not indicated as atopic dermatitis is only very rarely driven by food allergy. Diet restrictions, intended to eliminate exposure to allergenic foods, are not only ineffective but may increase stress unnecessarily.

Fingernails should be cut short to minimize excoriations and secondary infections.

Topical corticosteroids

Topical corticosteroids are the mainstay of therapy.

Creams or ointments applied 2 times a day are effective for most patients with mild or moderate disease. Low- to mid-potency corticosteroids often control skin inflammation, but high-potency corticosteroids are needed for resolution of chronically inflamed, lichenified skin. Dermatologic adverse effects of topical corticosteroids include thinning of the skin, striae distensae, and skin infections. Adverse effects vary and depend upon the potency of the corticosteroid, the duration of its use, and the skin location where it is applied. Prolonged, widespread use of high-potency corticosteroids should be avoided, particularly in infants, to avoid adrenal suppression. Corticosteroids should be stopped once the skin inflammation is controlled and should never be used to prevent recurrences.

Systemic corticosteroids can provide emergency relief, but long-term use should be avoided due to their multiple adverse effects. Because atopic dermatitis tends to flare when systemic corticosteroids are stopped, other treatments are typically required to manage tapering and discontinuation of systemic corticosteroids.

Topical calcineurin inhibitors

Topical tacrolimus and pimecrolimus are calcineurin inhibitors. They are T-cell inhibitors and can be used for mild to moderate atopic dermatitis or when corticosteroid adverse effects are a concern.

Tacrolimus ointment or pimecrolimus cream is applied 2 times a day.

Burning or stinging after application is usually transient and abates after a few days. Flushing is less common.

Topical crisaborole

Crisaborole is a topical phosphodiesterase-4 inhibitor. Crisaborole 2% ointment can be used for mild to moderate atopic dermatitis in patients 2 years of age or older.

Crisaborole is applied to areas of eczema 2 times a day. It cannot be used on mucous membranes.

Burning or stinging after application is the most common adverse effect.

Topical ruxolitinib

Ruxolitinib is a Janus kinase (JAK) inhibitor. Ruxolitinib 1.5% cream can be used for the short-term and noncontinuous treatment of mild to moderate atopic dermatitis in immunocompetent patients 12 years of age and older whose disease is not adequately controlled with other topical prescription therapies, or when those therapies are not advisable.

Ruxolitinib is applied as a thin layer 2 times a day to up to 20% of the body surface area for up to 8 weeks.

Phototherapy

Phototherapy with narrow-band ultraviolet B (UVB) is helpful for extensive atopic dermatitis, particularly when appropriate skin care and topical treatments fail to control inflammation. Because narrow-band UVB has greater efficacy than the previously used broad-band UVB, psoralen plus UVA (PUVA) therapy is rarely used. While an increase in skin cancer risk with narrow-band UVB phototherapy has not been demonstrated, this remains a potential concern, particularly when used in children or for extended periods of time. This risk needs to be weighed against risks of other systemic treatments after failure of appropriate skin care and topical treatments.

If office-based phototherapy is not available or is too inconvenient, home phototherapy is a good alternative. Several home phototherapy devices have programmable features that allow specialists to monitor and supervise a patient's use of the device.

Natural sun exposure is an alternative when phototherapy is not available.

Systemic immunosuppressants

Systemic immunosuppressants, such as cyclosporine, mycophenolate, methotrexate, and azathioprine, inhibit T-cell function. These medications are indicated for widespread, recalcitrant, or disabling atopic dermatitis that fails to abate with topical therapy and phototherapy. Their use must be balanced against risk of adverse effects, particularly with long-term use.

For moderate to severe atopic dermatitis that does not respond to topical therapies, first-line treatment options are the biologic agents dupilumab or tralokinumab.

Dupilumab is a fully human monoclonal IgG4 antibody that blocks the signaling of the proinflammatory Th2 cytokines IL-4 and IL-13 in atopic dermatitis. It is available for the treatment of moderate to severe atopic dermatitis in patients 6 years of age or older and is recommended for patients whose disease is not adequately controlled with other treatments.

Dupilumab is given subcutaneously in a regimen that involves a loading dose followed by periodic (weekly or biweekly) dosing. The most common adverse effects are injection site reactions, eye inflammation and irritation (conjunctivitis, blepharitis, keratitis, pruritus, and dry eye), herpes simplex virus infections, and eosinophilia.

Tralokinumab is a fully human monoclonal antibody against the IL-13 receptors alpha 1 and alpha 2. It is available for the treatment of moderate to severe atopic dermatitis in adults whose disease is not adequately controlled with topical therapies.

Tralokinumab-ldrm is also given subcutaneously. The most common adverse effects are upper respiratory infections, conjunctivitis, injection site reactions, and eosinophilia.

Oral JAK inhibitors (upadacitinib, abrocitinib, and baricitinib) are the newest systemic immunosuppressants that also may be used in patients 12 years of age or older with moderate to severe atopic dermatitis and inadequate disease control with other systemic medications, including biologic agents. Risks include serious infections, cancer, cardiovascular events, thrombosis, and death.

Treatment of superinfections

Antistaphylococcal antibiotics, both topical (eg, mupirocin, fusidic acid) and oral, are used to treat bacterial skin superinfections, such as impetigo, folliculitis, or furunculosis. *Staphylococcus aureus,* the most common bacterium causing skin infections in patients with atopic dermatitis, is often resistant to methicillin (methicillin-resistant *S. aureus* [MRSA]).

Doxycycline or trimethoprim/sulfamethoxazole are good initial choices for systemic antibiotics because MRSA is most often sensitive to these antibiotics. However, bacterial cultures with resistance testing are recommended before starting systemic antibiotics because resistance cannot be predicted.

Nasal mupirocin is recommended for patients who are nasal carriers of *Staphylococcus aureus*, a potential source of recurrent impetiginization. Screening for *Staphylococcus aureus* colonization (eg, of the nasal canals) should be considered in patients with frequent superinfections.

Eczema herpeticum is treated with systemic antivirals (eg, acyclovir, valacyclovir). Involvement of the eye is considered an ophthalmic emergency. If eye involvement is suspected, an ophthalmology consult should be obtained.

Treatment references

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Prognosis for Atopic Dermatitis

Atopic dermatitis in children often abates by age 5 years, although exacerbations are common throughout adolescence and into adulthood.

Girls and patients with severe disease, early age of onset, family history, and associated allergic rhinitis or asthma are more likely to have prolonged disease. Even in these patients, atopic dermatitis frequently resolves or lessens significantly by adulthood.

Atopic dermatitis may have long-term psychologic sequelae as children confront the challenges of living with a visible, sometimes disabling skin disease during their formative years.

Key Points

- **Atopic dermatitis is common, particularly in developed nations, affecting up to 20% of children and 10% of adults.**
- **A genetically determined skin barrier defect predisposes to inflammation with skin irritants and thus atopic dermatitis.**
- **Common triggers include excessive washing and bathing.**
- **Common findings vary with age and include pruritus and scaly erythematous patches and plaques and lichenification in the antecubital and popliteal fossae and on the eyelids, neck, and wrists.**
- **Superinfections (particularly *S. aureus* infections and eczema herpeticum) are common.**
- **Atopic dermatitis frequently resolves or lessens significantly by adulthood.**
- **First-line treatments include moisturizers, topical corticosteroids, and antihistamines as needed for pruritus.**

- For disease unresponsive to topical therapy, consider phototherapy or systemic immunosuppressants.

Drug Information for the Topic